User Stories

Project: RAG-Based Retrieval for Archival Systems
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Roles

Role	Description
Researcher	Academic or graduate student querying the archive for primary sources, historical data, or literature. Needs precise, citable answers.
Professional	Practitioner (e.g., lawyer, journalist, museum curator) who needs to quickly locate specific archival records or policies. Needs fast, trustworthy responses.
General Public User	Member of the public exploring archival content. Needs accessible, plain-language responses without domain expertise.
Archivist / Admin	Institutional staff managing the Islandora repository, approving ingestions, and maintaining collection quality.
System Administrator	Engineer responsible for deploying, monitoring, and maintaining the RAG infrastructure.

Epic 1: Natural Language Document Retrieval

Users can ask questions in natural language and receive grounded, cited answers from the archival collection.

US-01: Ask a Natural Language Question

As a researcher,
I want to type a question in natural language
so that I can find relevant archival information without learning a query syntax or search operators.

Priority: High
Effort: L

Acceptance Criteria:

- The chat interface accepts free-text input
- A response is returned within 10 seconds
- The response directly addresses the question's intent

US-02: Receive Source-Attributed Response

As a researcher,
I want every response to include citations naming the source document, section, and page
so that I can verify the answer independently and use it in academic work.

Priority: High
Effort: M

Acceptance Criteria:

- Each factual claim in the response is accompanied by a citation
- The citation identifies at minimum the document name and section/page
- Clicking or tapping a citation opens or highlights the source passage

US-03: Ask Follow-Up Questions

As a researcher,
I want to ask follow-up questions that reference my earlier query
so that I can explore a topic iteratively without restating context every time.

Priority: Medium
Effort: M

Acceptance Criteria:

- A follow-up such as "Can you elaborate on the second point?" resolves correctly to prior context
- Session history is maintained for the duration of the session
- The system does not mix context from different sessions

US-04: Query in My Own Language

As a general public user whose primary language is not English,
I want to submit queries in my native language (e.g., German)
so that I can access archival content without being fluent in the archive's primary language.

Priority: Medium
Effort: L

Acceptance Criteria:

- Queries in supported languages are correctly understood and answered
- Results from source documents in different languages are returned when relevant
- The response language matches the query language

US-05: Filter by Document Metadata

As a professional,
I want to narrow my search by date range, collection, or document type
so that I get results from the specific subset of the archive relevant to my task.

Priority: Medium
Effort: M

Acceptance Criteria:

- The interface exposes date range, collection tag, and document type filters
- Retrieval respects applied filters without the user embedding them in query text
- Filters can be combined

US-06: Get Results Despite Historical Terminology

As a researcher working with pre-20th century archives,
I want the system to find relevant documents even when I use modern vocabulary
so that vocabulary mismatch between my query and historical source language does not cause gaps in results.

Priority: High
Effort: L

Acceptance Criteria:

- A query for "aviation" returns documents containing "flying machines" where relevant
- Hybrid retrieval (dense + sparse) is active by default
- No user configuration is required to benefit from semantic bridging

US-07: Query from the Drupal Front-End

As a general public user,
I want to query the archive directly within the Drupal repository site
so that I can access institutional knowledge without leaving the platform I'm already browsing.

Priority: Medium
Effort: L

Acceptance Criteria:

- The Drupal front-end provides a chat interface for submitting queries
- Responses and citations render correctly within the Drupal page
- The front-end connects to the backend query flow orchestrated by Node-RED

Epic 2: Document Ingestion & Collection Management

Archivists can add new documents to the collection and have them become immediately queryable.

US-08: Ingest a New Document via Islandora

As an archivist,
I want to add a new digitised document to Islandora
so that it becomes searchable by all users without requiring developer intervention.

Priority: High
Effort: L

Acceptance Criteria:

- Uploading a document to Islandora triggers automatic pipeline processing
- The document's text is extracted, chunked, embedded, and indexed in ChromaDB
- The document is queryable within 15 minutes for a 100-page document

US-09: Preserve Document Structure During Ingestion

As an archivist,
I want document section headings, hierarchy, and metadata preserved in each chunk
so that retrieved passages can be correctly understood in context by the LLM.

Priority: High
Effort: M

Acceptance Criteria:

- Each stored chunk includes `context_heading`, `context_content`, `context_links`, and metadata fields
- No chunk boundary splits a sentence mid-phrase or separates a table row from its header
- Structure-aware chunking is used, not naive character-count splitting

US-10: Track Ingestion Status

As an archivist,
I want to see the ingestion status of each document I submit
so that I know when it is available for queries and can investigate failures.

Priority: Low
Effort: M

Acceptance Criteria:

- The admin interface shows per-document status: pending, processing, indexed, failed
- Failed ingestions display a human-readable error message
- Successful ingestions show a timestamp and chunk count

US-11: Remove a Document from the Index

As an archivist,
I want to remove a document from the retrieval index
so that users cannot receive responses grounded in outdated, incorrect, or restricted content.

Priority: Medium
Effort: M

Acceptance Criteria:

- Removing a document triggers deletion of its chunks from ChromaDB
- Subsequent queries no longer return content from that document
- The operation does not require a full re-index of the collection

Epic 3: System Operations

The system administrator can deploy and maintain the pipeline on a single GPU node.

US-12: Deploy on a Single 24 GB GPU

As a system administrator,
I want to run the full stack on a single NVIDIA L4 or RTX 4090
so that the prototype can be deployed without a multi-GPU cluster or cloud infrastructure.

Priority: High
Effort: L

Acceptance Criteria:

- LLM inference, embedding model, and reranker run within 24 GB VRAM
- VRAM usage is documented for the recommended model sizes (7B–14B quantized)
- RAGDoll-style offloading is configured to handle memory contention

US-13: Keep All Data On-Premises

As a system administrator,
I want all inference and document storage to remain on-premises
so that institutional data sovereignty and privacy requirements are met.

Priority: High
Effort: M

Acceptance Criteria:

- No document content, query text, or generated response is transmitted to an external API

- LLM runs via OLLAMA or vLLM on local hardware
- All dependencies are open-source with no mandatory cloud licensing